ECON 306 Problem Set 4

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Note: per the assignment instructions, the Stata log file goes first in the submitted PDF, followed by these written answers. The Stata code is reproduced in the appendix. All results below were computed from CAschool_PS4_AI.dta (420 districts) and double-checked.

1. Creating Variables

Income. The original variable was renamed income_original and left unchanged. The cleaned version strips commas and both suffixes found in the file (314 rows end in " dollars", 106 end in " dollars USD"), is converted to a number, and is divided by 1,000, because the textbook (chapter 8.1) measures district income in thousands of dollars. The scale is confirmed by section 2b, which reproduces the textbook equation (8.2) coefficients (607.3, 3.85, -0.04).

Students. Students is not stored directly, but str is the student-teacher ratio, that is, students per teacher, so total students equal teachers times str: students = teachers x str. (Teachers itself is stored as a string and was converted to a number first.) Check: computer/students reproduces the comp_stu variable to within 0.000001, so the construction is right.

Log and dummy variables. ln_income is the natural log of income (in thousands of dollars); ln_testscore is the natural log of testscore. math_lower equals 1 when a district's math score is below its reading score. Following textbook section 8.3, HiSTR equals 1 when the student-teacher ratio is 20 or more (182 districts) and HiEL equals 1 when the English-learner share is 10% or more (192 districts).

2. Replication

2b. Equation (8.2). R2 = 0.5562, adj. R2 = 0.5540.

Variable	Coef.	Robust SE	p-value
Constant	607.3017	2.9018	0.0000
income	3.8510	0.2681	0.0000
income2	-0.0423	0.0048	0.0000

Test scores rise with income but at a diminishing rate: the linear term is positive and the squared term is negative.

2c. Equation (8.11). R2 = 0.5584, adj. R2 = 0.5552.

Variable	Coef.	Robust SE	p-value
Constant	600.0790	5.1021	0.0000
income	5.0187	0.7074	0.0000
income2	-0.0958	0.0290	0.0009
income3	0.000685	0.0003	0.0483

The cubic term is significant at 5% but tiny, and R2 rises only from 0.5562 to 0.5584, so the cubic adds almost nothing over the quadratic.

2d. Equation (8.18). R2 = 0.5625, adj. R2 = 0.5615.

Variable	Coef.	Robust SE	p-value
Constant	557.8323	3.8399	0.0000
ln_income	36.4197	1.3969	0.0000

A 1% increase in income is associated with a 36.4197/100 = 0.364 point rise in test scores.

2e. Equation (8.23). R2 = 0.5578, adj. R2 = 0.5567.

Variable	Coef.	Robust SE	p-value
Constant	6.3363	0.0059	0.0000
ln_income	0.0554	0.0021	0.0000

The log-log slope is an elasticity: a 1% increase in income is associated with a 0.055% increase in test scores.

2f. Equation (8.30). R2 = 0.2956, adj. R2 = 0.2905.

Variable	Coef.	Robust SE	p-value
Constant	664.1433	1.3881	0.0000
HiSTR	-1.9078	1.9322	0.3235
HiEL	-18.1629	2.3460	0.0000
HiSTR_HiEL	-3.4943	3.1212	0.2629

Effect of moving from a high student-teacher ratio to a low one. The answer depends on HiEL because of the interaction term. When HiEL = 0, the effect is -(-1.9078) = +1.9078 points. When HiEL = 1, the effect is -(-1.9078 + -3.4943) = +5.4022 points. Neither HiSTR nor the interaction is individually significant at 5%, so these effects are imprecisely estimated, while the HiEL gap itself (-18.1629 points) is large and significant.

2a. Scatterplot from Figure 8.3

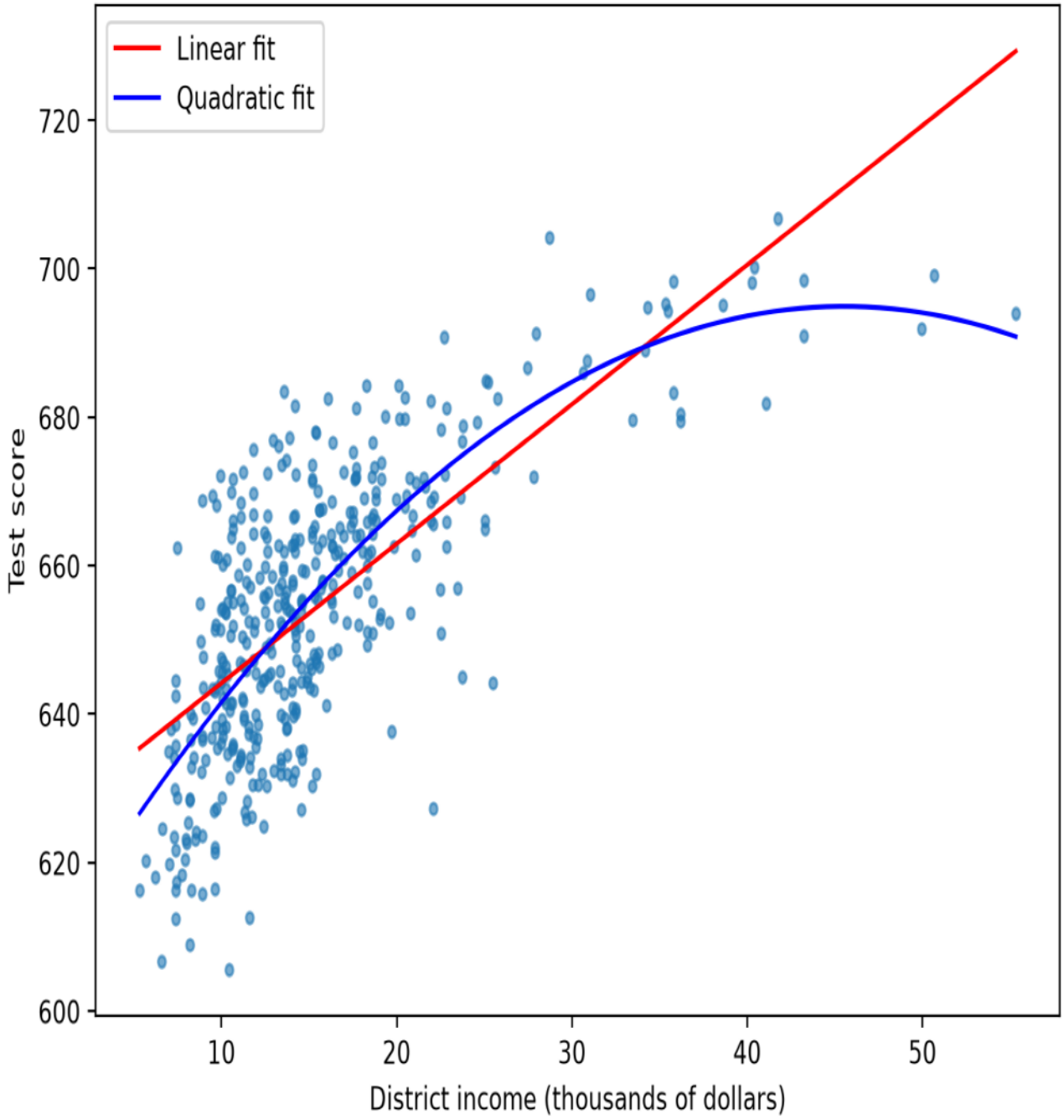


Figure 8.3 replication: test scores against district income (thousands of dollars), with the linear fit (red) and the quadratic fit (blue). The quadratic tracks the flattening of the relationship at higher incomes, matching the negative squared term in equation (8.2).

3. Measurement Error

3a. testscore on students and income. R2 = 0.5379.

Variable	Coef.	Robust SE	p-value
Constant	627.4154	1.9720	0.0000
students	-0.000849	0.0001	0.0000
income	1.8916	0.1187	0.0000

Slope on students: 1,000 more students are associated with 0.849 fewer test score points, holding income fixed.

3b. Same regression with noisy_students. R2 = 0.5346.

Variable	Coef.	Robust SE	p-value
Constant	627.3538	1.9751	0.0000
noisy_students	-0.000782	0.0001	0.0000
income	1.8886	0.1182	0.0000

Almost nothing changes. Classical measurement error in X biases the slope toward zero, but here the added noise is small relative to the variation in students (noise standard deviation 740 against a students standard deviation of 3,913; correlation between students and noisy_students is 0.9828), so the attenuation is barely visible: -0.000782 against -0.000849.

3c. Noisy testscore on noisy_students and income. R2 = 0.1173.

Variable	Coef.	Robust SE	p-value
Constant	630.1885	4.1838	0.0000
noisy_students	-0.000375	0.0004	0.3067
income	1.5853	0.2706	0.0000

Adding noise to Y changes the picture sharply. Measurement error in Y does not bias the slopes, but it inflates their standard errors and destroys fit: the students slope halves and loses significance, the income standard error more than doubles (0.2706 against 0.1182), and R2 collapses from 0.5346 to 0.1173. That is expected here because the test-score noise (standard deviation 29.6) is larger than the standard deviation of test scores themselves (19.1).

4. Logarithms

4a. testscore on students and ln(income). R2 = 0.6049.

Variable	Coef.	Robust SE	p-value
Constant	558.6545	3.8319	0.0000
students	-0.001005	0.0001	0.0000
ln_income	37.1080	1.3770	0.0000

Logged income: a 1% increase in income is associated with 37.1080/100 = 0.371 more test score points. Non-logged income (section 3a): a $1,000 increase in income is associated with 1.8916 more points. The two are not directly comparable because one is a percentage change and the other is a dollar change; the log form fits better (R2 0.6049 against 0.5379).

4b. Log-log (continued)

4b. ln(testscore) on ln(students) and ln(income). R2 = 0.6135.

Variable	Coef.	Robust SE	p-value
Constant	6.3647	0.0074	0.0000
ln_students	-0.005053	0.0006	0.0000
ln_income	0.0581	0.0021	0.0000

Both slopes are elasticities. A 1% increase in students is associated with a 0.0051% decrease in test scores, and a 1% increase in income with a 0.0581% increase in test scores.

5. F Test in Stata

Full regression. R2 = 0.7407, adj. R2 = 0.7350.

Variable	Coef.	Robust SE	p-value
Constant	631.7330	7.2255	0.0000
students	0.000017	0.0011	0.9876
income	2.3475	0.6453	0.0003
income2	-0.004061	0.0257	0.8745
income3	-0.000324	0.0003	0.2743
math_lower	-4.7964	1.0605	0.0000
teachers	-0.004508	0.0228	0.8431
expn_stu	-0.000339	0.0009	0.7048
comp_stu	17.9574	8.5937	0.0367
el_pct	-0.4743	0.0349	0.0000

Five coefficients are not statistically significant at 5%: students, income2, income3, teachers, and expn_stu. The F test sets all five equal to zero. $F(5, 410) = 6.9344$, $p = 0.000003$. (The heteroskedasticity-robust Wald version of the same test gives $F = 17.1442$, $p < 0.000001$.) Both are far above the 1% critical value of 3.0621, so the null is rejected: the five variables are jointly significant even though none is individually significant. That is the signature of multicollinearity: income, income2, and income3 move together, and students, teachers, and the student-teacher ratio are mechanically linked, so the individual t-tests cannot separate their contributions while the joint test can. Dropping all five on the basis of the individual t-tests would be a mistake.

6. F Test by Hand (alpha = 0.01)

The null hypothesis is the same as in section 5: the coefficients on students, income2, income3, teachers, and expn_stu are all zero. The test uses the residual sums of squares from the restricted regression (those five variables dropped) and the unrestricted regression (all variables included).

Unrestricted residual sum of squares (RSSu) = 39,439.51
Restricted residual sum of squares (RSSr) = 42,774.73
Number of restrictions q = 5
Denominator degrees of freedom = n - k = 420 - 10 = 410

F = ((RSSr - RSSu) / q) / (RSSu / (n - k))
  = ((42,774.73 - 39,439.51) / 5) / (39,439.51 / 410)
  = (3,335.22 / 5) / 96.19
  = 667.04 / 96.19
  = 6.9344

Critical value: F(5, 410) at the 1% level = 3.0621.

6.9344 > 3.0621, so the null is rejected at the 1% level. The conclusion matches section 5 exactly (the by-hand value equals the Stata value up to rounding): the five individually insignificant variables are jointly significant and should not all be dropped.

Appendix: Stata Code (PS4-CORRECTED.do)

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* ECON 306 Problem Set 4
* AI used: Muse, for all sections (variable creation, replication, measurement
* error, logarithms, and both F-tests). Each AI-assisted section is marked below.
* Data: CAschool_PS4_AI.dta (420 districts)

log using PS4, replace

use CAschool_PS4_AI.dta, clear

* Section 1: Creating Variables
*AI helped
rename income income_original
gen income_str = income_original
replace income_str = subinstr(income_str, ",", "", .)
replace income_str = subinstr(income_str, " dollars USD", "", .)
replace income_str = subinstr(income_str, " dollars", "", .)
destring income_str, replace
gen income = income_str / 1000
drop income_str
gen income2 = income^2
gen income3 = income^3
gen ln_income = ln(income)
gen math_lower = (math_scr < read_scr)
* students is not stored directly. str is the student-teacher ratio, students
* per teacher, so students = teachers x str.
destring teachers, replace
gen students = teachers * str
gen ln_testscore = ln(testscore)
gen HiSTR = (str >= 20)
gen HiEL = (el_pct >= 10)
*AI helped

* Section 2: Replication
*AI helped
twoway scatter testscore income || lfit testscore income || qfit testscore income
*AI helped
regress testscore income income2, robust
display e(r2)
display e(r2_a)
* 8.2: 607.3017 + 3.8510 income - 0.0423 income2, R2 = 0.5562
*AI helped
regress testscore income income2 income3, robust
display e(r2)
display e(r2_a)
* 8.11: 600.0790 + 5.0187 income - 0.0958 income2 + 0.000685 income3, R2 = 0.5584
*AI helped
regress testscore ln_income, robust
display e(r2)
display e(r2_a)
* 8.18: 557.8323 + 36.4197 ln_income, R2 = 0.5625
*AI helped
regress ln_testscore ln_income, robust
display e(r2)
display e(r2_a)
* 8.23: 6.3363 + 0.0554 ln_income, R2 = 0.5578
*AI helped
gen HiSTR_HiEL = HiSTR * HiEL
regress testscore HiSTR HiEL HiSTR_HiEL, robust
display e(r2)
display e(r2_a)
* 8.30: 664.1433 - 1.9078 HiSTR - 18.1629 HiEL - 3.4943 HiSTR_HiEL, R2 = 0.2956
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*AI helped

* Section 3: Measurement Error
*AI helped
regress testscore students income, robust
display e(r2)
display e(r2_a)
* 3a: 627.4154 - 0.000849 students + 1.8916 income, R2 = 0.5379
*AI helped
regress testscore noisy_students income, robust
display e(r2)
display e(r2_a)
* 3b: 627.3538 - 0.000782 noisy_students + 1.8886 income, R2 = 0.5346
*AI helped
regress noisy_testscore noisy_students income, robust
display e(r2)
display e(r2_a)
* 3c: 630.1885 - 0.000375 noisy_students + 1.5853 income, R2 = 0.1173
*AI helped

* Section 4: Logarithms
*AI helped
regress testscore students ln_income, robust
display e(r2)
display e(r2_a)
* 4a: 558.6545 - 0.001005 students + 37.1080 ln_income, R2 = 0.6049
*AI helped
gen ln_students = ln(students)
regress ln_testscore ln_students ln_income, robust
display e(r2)
display e(r2_a)
* 4b: 6.3647 - 0.005053 ln_students + 0.0581 ln_income, R2 = 0.6135
*AI helped

* Section 5: F test in Stata
*AI helped
* Classical standard errors here so the Stata F matches the by-hand F in section 6.
regress testscore students income income2 income3 math_lower teachers expn_stu comp_stu el_pct
test students income2 income3 teachers expn_stu
* F(5, 410) = 6.9344, p = 0.000003: reject. The 5 variables are jointly significant.
*AI helped

* Section 6: F test by hand, alpha = 0.01
*AI helped
regress testscore math_lower income comp_stu el_pct
scalar rss_r = e(rss)
regress testscore students income income2 income3 math_lower teachers expn_stu comp_stu el_pct
scalar rss_u = e(rss)
scalar F = ((rss_r - rss_u) / 5) / (rss_u / 410)
display "RSS unrestricted = " rss_u
display "RSS restricted = " rss_r
display "F statistic = " F
display "Critical value at 1% = " invF(5, 410, 0.99)
* RSSu = 39439.51, RSSr = 42774.73, F = 6.9344, critical value = 3.0621: reject.
*AI helped

log close
translate PS4.smcl PS4.pdf
```